



Parallel disassembly sequence planning using improved ant colony algorithm

Yufei Xing^{1,2} · Dongmei Wu³ · Ligang Qu^{1,2}

Received: 7 September 2020 / Accepted: 1 February 2021 / Published online: 21 February 2021
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Abstract

Disassembly is one of the most crucial links in the product life cycle. Disassembly sequence planning (DSP) is a combination of optimization problem and has been studied by many researchers. Asynchronous parallel disassembly planning (APDP) eliminates the requirement that manipulators must start and stop their tasks in the working process. In this paper, a new asynchronous parallel disassembly sequence planning method is proposed, which to deal with the problems of highly time-dependent on asynchronous parallel disassembly mode. The disassembly time model has been divided into execution time and preparation time. The former represents the time consumes by removing a part, and the latter means the time consumes on the disassembly tool preparation while the disassembly position switch. The disassembly matrix and the disassembly work area collision matrix are also utilized to avoid infeasible sequences. Based on the max-min ant system, a waiting strategy is proposed to solve asynchronous parallel disassembly sequence planning problem. To further shorten disassembly process, a time overlap strategy is developed to take advantage of waiting time. Finally, a bevel gear reducer is utilized as an example to discuss and analysis the value of key parameters, as well as the effectiveness of the algorithm and strategies proposed in the paper.

Keywords Combinatorial optimization · Disassembly sequence planning (DSP) · Parallel disassembly · Disassembly time-dependent · Ant colony algorithm

1 Introduction

In the past ten years, due to rapid development of products and needs of resource recycling, much attentions have been given to deal with end-of-life products. How to perform these EOL activities efficiently and economically is a significant research problem. Disassembly is a process of separating a product into its constituent parts, which subassemblies according to a certain logical order and geometric constraint relationship, and has solved problem of reuse, recycling, or remanufacture of EOL products [1–3].

Disassembly sequence planning (DSP) determines the best order that components are removed from products aiming at minimizing the disassembly time, cost, or other factors while considering the disassembly direction and other attributes of components. That can be viewed as a challenging NP-hard combinatorial optimization problem. It is necessary to find a feasible and optimal disassembly sequence. An optimal or near optimal disassembly sequence could maximize economic and environmental benefits [4]. Due to the disassembly sequence that can drastically improve the efficiency of disassembly process, the DSP problem has been studied widely. Disassembly sequence planning consists of three major steps, namely, disassembly mode deciding, disassembly model construction, and disassembly sequence planning [3].

According to the number of manipulators participated in disassembly, whose mode can be divided into two categories: sequential disassembly mode and parallel disassembly mode. For sequential disassembly mode, all the disassembly tasks are implemented by one manipulator, and all the parts are removed one by one. For parallel disassembly mode, multiple manipulators remove parts simultaneously. Therefore, DSP can be classified into sequential disassembly sequence

✉ Yufei Xing
2371915729@qq.com

¹ College of Mechanical and Electrical Engineering, Shenyang Aerospace University, Shenyang 110136, Liaoning, China

² Key Laboratory of Fundamental Science for National Defense of Aeronautical Digital Manufacturing Process, Shenyang Aerospace University, Shenyang 110136, Liaoning, China

³ School of Business Administration, Northeastern University, Shenyang 110819, Liaoning, China

planning (SDSP) and parallel disassembly sequence planning (PDSP). In SDSP, only one part or subassembly is removed one at a time [5–10], which follows a linear process. Thus, the sequential disassembly may lead to a long processing time, especially for large or complex products [11–15]. In contrast to SDSP, the part or subassembly can be disassembled by different manipulators at the same time in PDSP, which could reduce disassembly time [16].

The disassembly sequencing is represented by graph [17]. The graph-based method [16, 18–21], Petri net [22, 23], and assembly/disassembly matrix [8, 24–29] have been proposed successively to represent the disassembly relation of parts in the product and guarantee the feasibility of the disassembly sequence. For PDSP problem, Kang [16] developed an action-oriented disassembly graph, called the extended process graph. The results was obtained by transforming the AND/OR graph, which could give the optimal disassembly level and sequence at the same time while considering the sequence-dependent operation times. Hung [11] presented selective parallel disassembly planning methods for green design. Based upon atomic theory, parts were grouped into independent modules. The disassembly sequence structure graphs (DSSG) were utilized to remove target parts from target modules. Tao [30] considered partial and parallel disassembly sequence planning for solving the degrees of freedom in modular product design. An automatic self-decomposed disassembly precedence matrix (DPM) was designed to generate partial/parallel disassembly sequence for reducing complexity and improving efficiency. Zhang [31] investigated the problem of parallel disassembly with consideration of fuzziness. Disassembly time was characterized by the fuzzy sets due to inevitable uncertainties. Further, in the parallel disassembly problem, the activities of manipulators were taken into account. In general, those manipulators are working separately, and the sub-disassembly sequences are parallel but asynchronous, which allows manipulators to start their tasks immediately without waiting others, as long as precedence (and other) constraints are not violated. Thus, it makes the PDSP become highly disassembly time-dependent [12, 15] and space-dependent [12] at the same time. To deal with this

problem, Ren [12] proposed asynchronous parallel disassembly planning that eliminated the synchronization requirement, and the work area collisions were also considered. For a large or complex product disassembly task, the cooperation between manipulators should be taken into account to minimize the total work time [14, 32]. Moreover, it allowed a part that can be disassembled cooperatively by more than one manipulator, as long as disassembly precedence and work area collisions were not occurred.

Compared to the SDSP problem, there are much fewer attentions that have been given to the solution of parallel disassembly sequence problem (PDSP) [3]. Dutta [33] presented a method of removing multiple components from an assembly based on a disassembly tree. Chen [34] presented the onion peeling algorithm for parallel disassembly by designing a mating graph. In order to find the near optimal solution in a reasonable computation time, a PDSP method based on fuzzy-rough sets has been presented by Zhang [31]. Based on the idea that the heuristic method was an effective way to solve NP-hard problems with both assembly sequence planning (ASP) [35–44] and SDSP problem [1, 5, 22, 45, 46], Ren [12, 47] designed an efficient encoding and decoding strategy for the asynchronous PDSP problem by using genetic algorithm. According to the uncertain characteristics of disassembly sequence length and the number of parts removed at each step for the PDSP, a chromosome coding method for parallel sequence was presented considering the disassembly sequence and disassembly steps simultaneously [48]. The branch-and-bound algorithm and bee colony algorithm were also addressed in the paper [13, 14]. A novel two-stage many-objective algorithm, which encoded parallel disassembly plans by using third-order tensors, was proposed [49]. To obtain the optimal multi-player cooperative disassembly sequence, the chromosome evolution rules was redesigned [50].

Analyzing the current states of the art, the few studies are reported on PDSP than SDSP. There are still two crucial issues that need to be dealt with.

First is the description of disassembly time. Since the PDSP problem highly depends on disassembly time, it is

Table 1 Disassembly time of each part given disassembly matrix P

Part index	Manipulator no.1		Manipulator no.2	
	Executing time	Preparation time	Executing time	Preparation time
1	20	10	20	10
2	20	20	20	20
3	20	20	20	30
4	25	10	25	10

crucial to consider the subdivision of the disassembly time based on the different action during the process of disassembly that shortens the total time. In previous research, only a few of them have considered the subdivision of disassembly time. To the best of our knowledge, the calculation method to deal with the subdivided disassembly time has not been proposed. In addition, the waiting time of asynchronous parallel disassembly also needs to be utilized.

Second, the parallel disassembly involves more than one manipulator; therefore, the construction of feasible solution becomes more complicated. The ant colony algorithm (ACA) is one of the most effective algorithms for solving discrete problems and widely used to solve both ASP [35, 38, 39, 51] and SDSP [5–7, 20, 25, 52–56] problems. However, less attention have been given to address the PDSP problems with the (ACA). However, the tradition sequential search strategy of the ant system is no longer applicable to deal with asynchronous PDSP problem.

Motivated by these two issues mentioned above, a new disassembly time calculation method to reduce the total disassembly time of the parallel solution by comparing and overlapping the subdivided disassembly time is proposed. Furthermore, based on the max-min ant system (MMAS) [5, 57], a novel algorithm is proposed to deal with the PDSP problem. A new search principle of the ant system with waiting strategy is designed to solve the asynchronous parallel disassembly sequence planning problem.

The rest of this paper is organized as follows: Section 2 give an overview of the new subdivide disassembly time model for PDSP and the precedence relationships of the PDSP process. Section 3 proposes an algorithm based on a max-min ant system (MMAS) to acquire the parallel disassembly sequence. Section 4 shows the computational results obtained from the methods proposed in the literature. Finally, Section 5 concludes the current study.

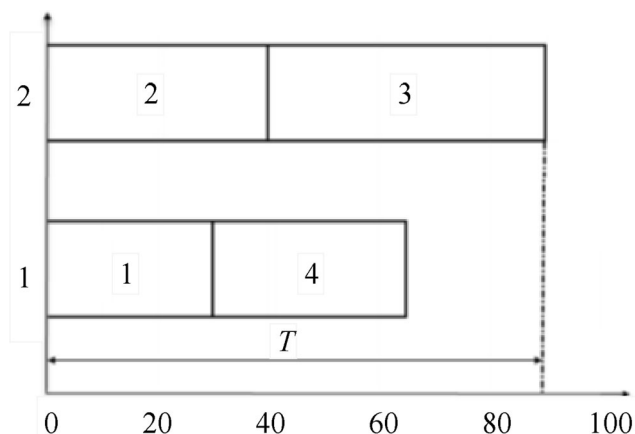


Fig. 1 Parallel disassembly without waiting strategy

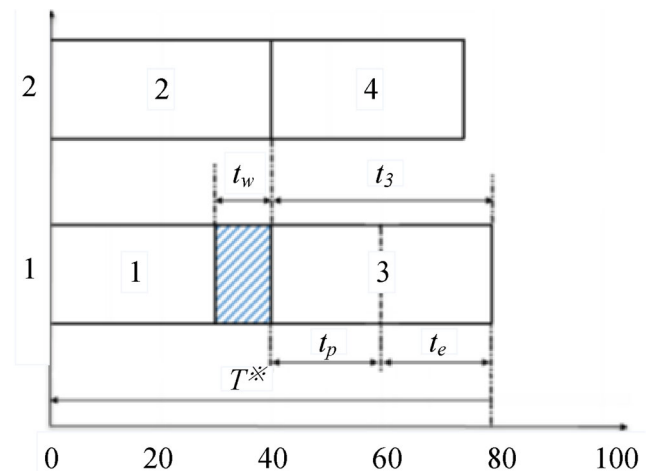


Fig. 2 Parallel disassembly with a waiting strategy

2 Problem statement

2.1 Problem representation

Here, a simple assembly contains four parts as a sample is used as an example to illustrate the difference between PDSP with and without waiting strategy. The disassembly precedence relationship of the example can be described by a disassembly matrix P based on the previous method of Xing et al. [26]. The disassembly time has been divided into execution time t_e which represents the time consumed by removing a part and preparation time t_p . Table 1 shows the disassembly time that consumes by preparing for disassembly. It is noteworthy that for different manipulators, the preparation time of the same part may not be equal due to differences in disassembly condition (such as disassembly tools and positions). Form the matrix P , for instance, parts 1, part 2, and part 4 are free to remove at the first place, and part 3 is blocked by part 2, indicating

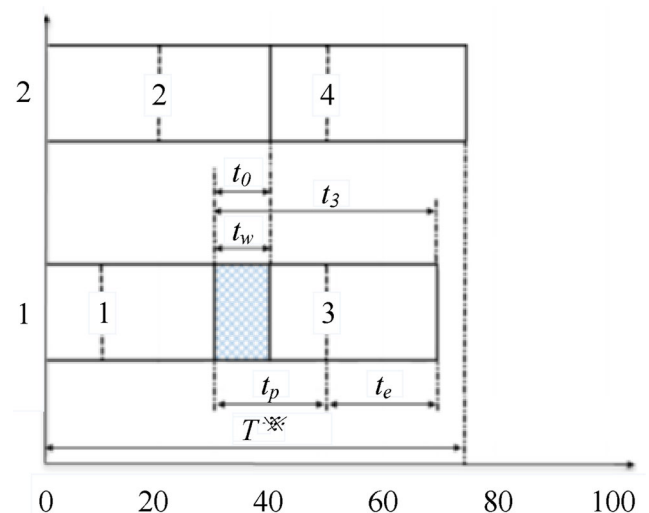


Fig. 3 Disassembly time overlap

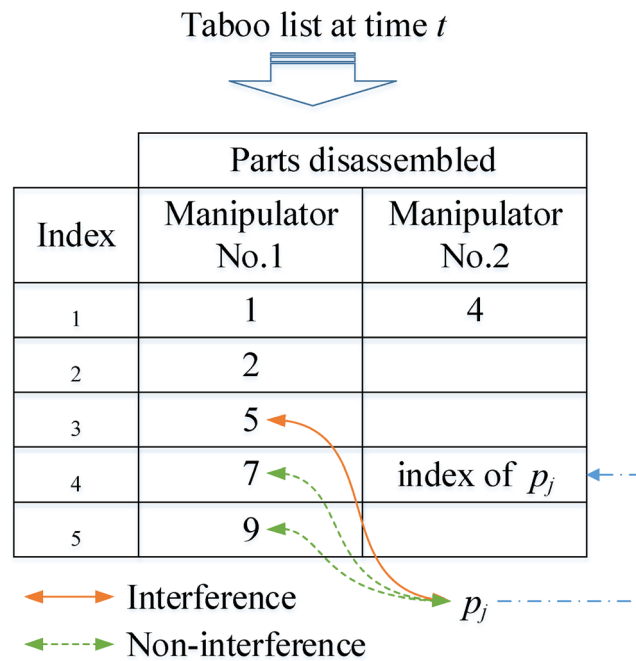


Fig. 4 An example of building the taboo list

that part 3 has to be disassembled after part 2. The parallel disassembly sequence without waiting strategy and with waiting strategy can be shown in Figs. 1 and 2, respectively.

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

In Fig. 1, part 1 and part 2 are disassembled by manipulator no.1 and no.2, respectively. Based on Table 1, the disassembly time of part 1 is shorter than part 2. Thus, it gives the manipulator no.1 the priority of choosing the next part to disassemble. However, based on the disassembly matrix P , part 4 was the only available part at that moment. Thus, the total disassembly time T is 90 s; on the contrary, in Fig. 2, instead of choosing the next part immediately after manipulator no.1, wait t_w seconds after removing part 1. Meanwhile, part 2 has been disassembled by manipulator no.2, and part 3 has no longer blocks. Therefore, it made the manipulator no.1 the chance to choose part 3. As mentioned above, manipulator no.2 spends longer preparation time than no.2 to remove part 3. Although, the manipulator No.1 spends time on wait but still shorter than the time taken by manipulator No.2 to remove part 3. The total disassembly time is shortened to T' and that is the reason that the total disassembly time in Fig. 2 is shorter than the solution given in

Table 2 Disassembly time of the study case

Disassembly time(s)			Execution time(s)			Preparation time(s)			Waiting time(s)		
No. 1	No. 2	No. 3	No. 1	No. 2	No. 3	No. 1	No. 2	No. 3	No. 1	No. 2	No. 3
9.70	8.20	8.90	9.70	8.20	8.90	----	----	----	----	----	----
9.84	8.32	9.02	9.70	8.20	8.90	0.14	0.12	0.12	0.00	0.00	0.00
9.84	8.32	9.02	9.70	8.20	8.90	0.14	0.12	0.12	0.00	0.00	0.00
9.84	8.32	9.02	9.70	8.20	8.90	0.14	0.12	0.12	0.00	0.00	0.00
12.04	7.50	12.50	7.50	6.00	11.00	4.54	1.50	1.50	0.00	0.00	0.00
7.60	55.00	55.00	7.50	40.00	40.00	0.10	15.00	15.00	0.00	0.00	0.00
7.60	15.00	0.00	7.50	13.00	0.00	0.10	2.00	0.00	0.00	0.00	0.00
7.60	7.00	0.00	7.50	4.00	0.00	0.10	3.00	0.00	0.00	0.00	0.00
10.50	17.00	50.20	9.00	15.00	0.00	1.50	2.00	0.00	0.00	0.00	0.00
13.53	0.00	40.00	5.30	0.00	40.00	8.23	0.00	0.60	0.00	0.00	50.20
5.42	0.00	0.00	5.30	0.00	0.00	0.12	0.00	0.00	0.00	0.00	0.00
5.30	0.00	0.00	5.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5.42	0.00	0.00	5.30	0.00	0.00	0.12	0.00	0.00	0.00	0.00	0.00
7.50	0.00	0.00	6.00	0.00	0.00	1.50	0.00	0.00	0.00	0.00	0.00
19.00	0.00	0.00	19.00	0.00	0.00	10.00	0.00	0.00	12.91	0.00	0.00
12.01	0.00	0.00	4.00	0.00	0.00	8.01	0.00	0.00	0.00	0.00	0.00
15.00	0.00	0.00	13.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00
11.00	0.00	15.01	8.00	0.00	0.00	3.00	0.00	0.00	0.00	0.00	0.00
17.00	0.00	40.00	15.00	0.00	40.00	2.00	0.00	0.00	1.99	0.00	15.01
10.00	0.00	0.00	10.00	0.00	0.00	5.50	0.00	0.00	40.00	0.00	0.00

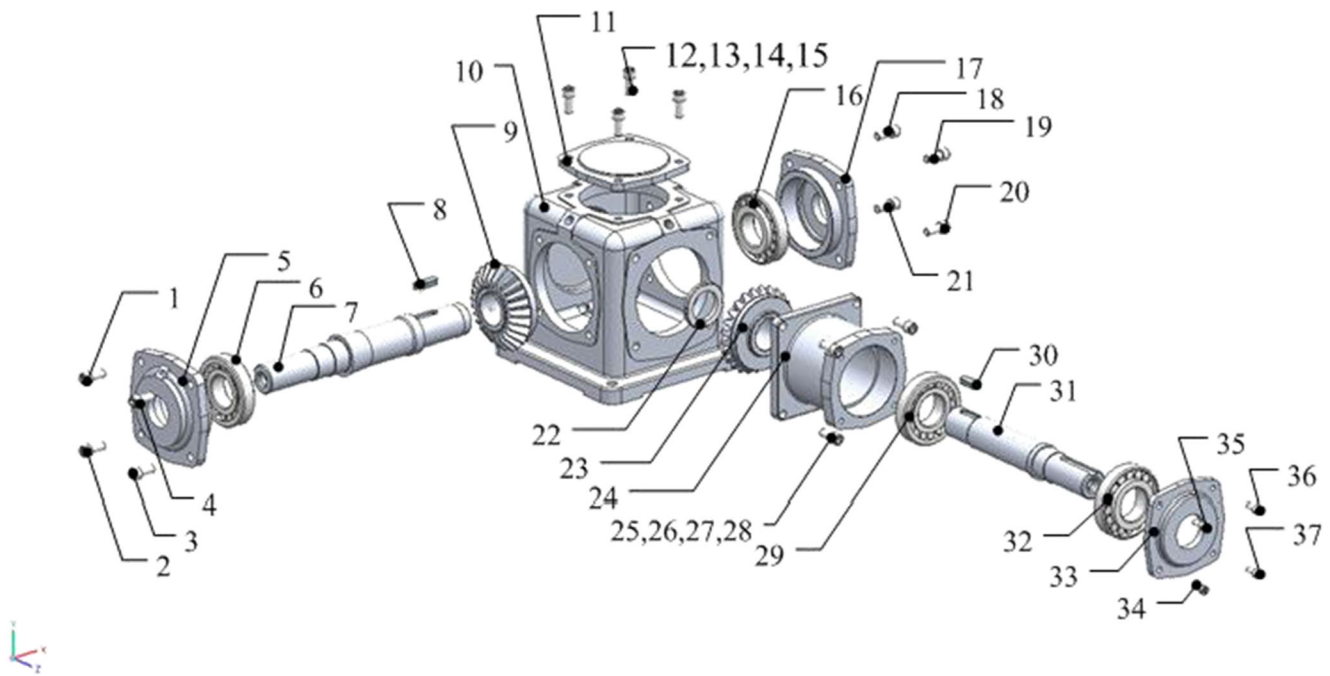


Fig. 5 Gear reducer assembly

Fig. 1. Therefore, disassembly time could be shortened when the waiting strategy is introduced into the PDSP problem.

Furthermore, when the manipulator no.1 chooses to wait, the waiting time t_w can be used for manipulator no.1 to prepare for the next disassembly part. As shown in Fig. 3, t_3 denotes the disassembly time of part 3, which is divided into t_e and t_p . Instead of preparing for disassembling part 3 after waiting t_w seconds, the manipulator no.1 starts preparing for the disassembly task of part 3 immediately, and t_0 denotes the overlap time. Due to the overlap of waiting time t_w and preparation time t_p , the total disassembly time is further shortened to T'' . Therefore, constructing a feasible disassembly sequence with a waiting mechanism is the key to overcome of this paper. The specific illustration is given in Section 3.

2.2 Mathematical model

To obtain the feasible parallel disassembly sequence, the disassembly precedence relationship and disassembly work area collision must be considered. The disassembly precedence between the parts can be guaranteed from the disassembly matrix (DM) that is given in [26]. P denotes the product that needs to be disassembled, which contains n parts. Let p_i and p_j represent part i and part j in P , respectively.

$$c_{dij} = \begin{cases} 1, & p_i \text{ interference with } p_j \text{ on } d \text{ direction} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The disassembly work area is defined as the space required for the manipulators participating in the disassembly to perform the disassembly task. During the disassembly process,

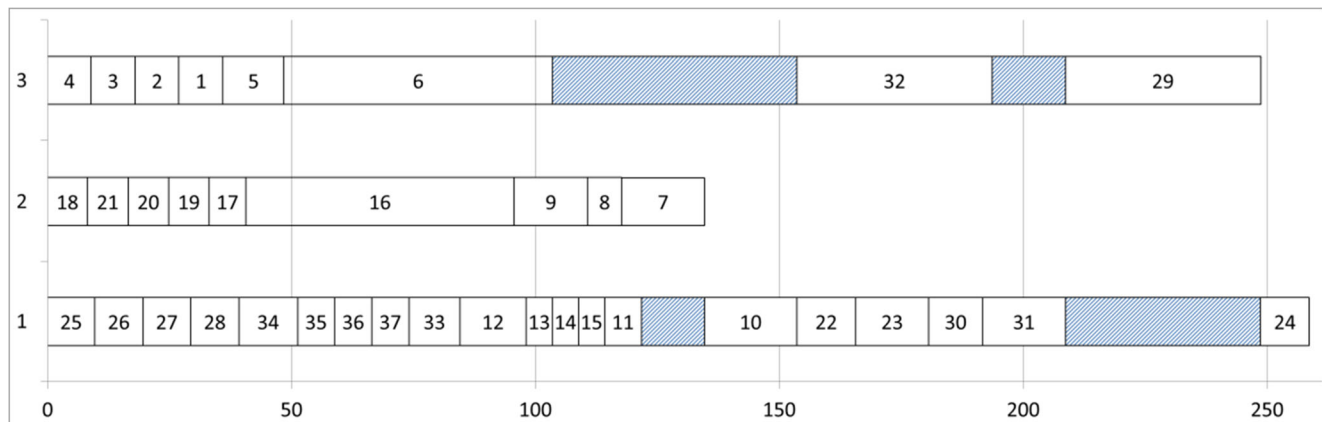


Fig. 6 Gantt chart of the best solution

Table 3 The result acquired by the proposed method and the method without overlap strategy

Method	Average(s)	Best(s)	Worst(s)
Proposed method	272.1372	258.6640	279.8163
Without overlap	284.9488	276.7469	294.5386

every manipulator needs to occupy a space of disassembly operation. The collision of the work area will inevitably lead to the disassembly task, not proceeding according to the planning. The interference relationship of the work area space can be obtained by work area collision matrix (WACM) proposed by Ren [12].

$$o_{ij} = \begin{cases} 1, & \text{work area collision between } p_i \text{ and } p_j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

For the characteristics of the parallel disassembly sequence planning problem, we divide the disassembly time into three parts: execution time, preparation time, and waiting time. Let $k(k = 1, 2, 3, \dots, n)$ denotes the manipulators that participate in the parallel disassembly process. The disassembly time of manipulator k depends on the disassembled part p_i that can be calculated as follows:

$$t_{k,j} = \sigma_{k,j} + \max\{\delta_{k,ij}, \varepsilon_{k,ij}\} \quad (3)$$

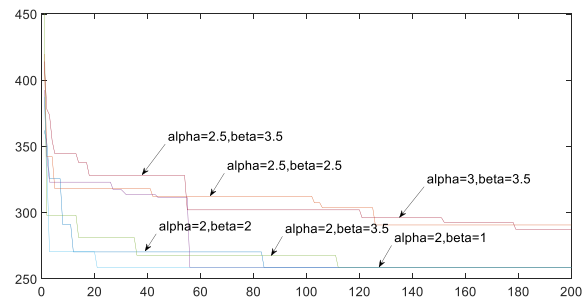
$\sigma_{k,i}$ denotes the execute time, which means the time taken to remove the part join constraints, and all the manipulators are assumed that spend the same time for the same part. The preparation time $\varepsilon_{k,ij}$ can be further subdivided into tool preparation time $\lambda_{k,ij}$ and disassembly position switching time $\theta_{k,ij}$.

The tool preparation time $\lambda_{k,ij}$ denotes the time taken by the manipulator k to disassemble the part p_j that required to prepare the corresponding disassembly equipment, which is related to the tools used for disassembling the previous part p_i .

$$\lambda_{k,ij} = \begin{cases} 0, & \text{same tool as previous part} \\ \lambda_{k,i}, & \text{else} \end{cases} \quad (4)$$

Table 4 Result comparison between the ACO and the other two methods

Method	Product	The number of parts	The number of manipulators	Average(s)	Best(s)	Worst(s)
Proposed method	An is takene block taken from [31]	35	2	1381	1270	1590
	A valve taken from [12]	40	3	349.5	338.2	367.2
The improved GA method proposed by [12]	An engine block is taken from [31]	35	2	1728	1731.9	1736
	A valve is taken from [12]	40	3	350.5	342	357

**Fig. 7** The graph of parameter analyze

$$\theta_{k,ij} = \begin{cases} \frac{d_{ij}}{V_k}, & \text{no contact between } p_i \text{ and } p_j \\ 0, & \text{else} \end{cases} \quad (5)$$

Let $\theta_{k,ij}$ denote the disassembly position switch time, and d_{ij} denotes the distance to cover if the manipulator k needs to move to next p_j , after disassembling the p_i . Let V_k denote the moving speed of the manipulators. d_{ij} represents the Manhattan distance between the bounding box center of p_i and p_j . In particular, if two parts are in contact, the time of position switch is set to 0. The contact relationship between parts can be acquired by the method given by Xing [26].

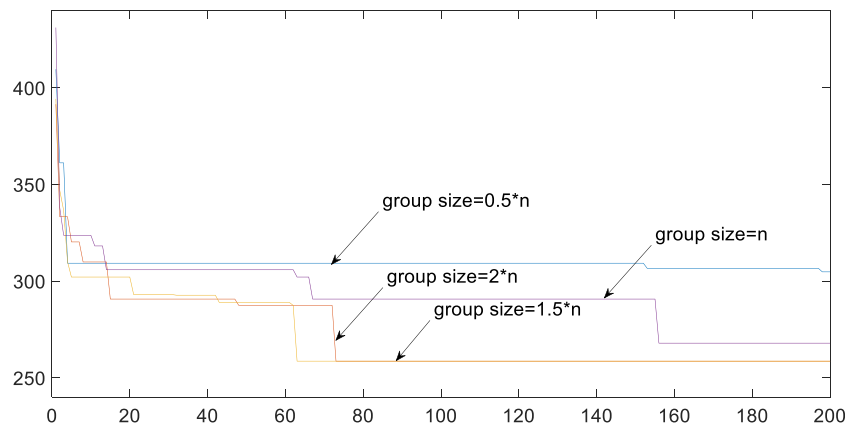
$\delta_{k,ij}$ denotes the time that the manipulator k disassembly p_j needs to wait for p_i to complete disassembly. It only takes place when p_i interference with p_j . The process of calculating waiting time is given in Section 3.2.

Let m denotes the number of disassembly tasks performed by manipulator k during the entire disassembly process, and then the total disassembly time of manipulator k can be calculated by Formula (6):

$$f(t_k) = \sum_{i=1}^m (t_{k,i}) \quad (6)$$

It is noteworthy that the disassembly time of parallel disassembly depends on the longest total disassembly time among the manipulators. Therefore, the disassembly time of parallel disassembly can be denoted by Formula (7).

Fig. 8 The graph of ant group size analyze



$$f(t) = \max\{f(t_k)\} \quad (7)$$

The disassembly time of parallel disassembly sequence is used as an optimization objective. The objective optimization function of parallel disassembly sequence can be obtained in Formula (8):

$$\min E[f(t)] = \min \left(\max \left\{ \sum_{i \in op_1} t_{1,i}, \sum_{i \in op_2} t_{2,i}, \sum_{i \in op_3} t_{3,i}, \dots, \sum_{i \in op_l} t_{k,i}, \dots, \sum_{i \in op_k} t_{K,i} \right\} \right) \quad (8)$$

3 Proposed algorithm

The ant colony optimization (ACO) is developed based on the behavior of the ants which work collectively to search and identify the nearest track from the nest to the food source. When moving along the track, the ants move from one position to another guided by a chemical substance known as pheromone. The ants will deposit a certain amount of pheromone on the path depending on the quality of the followed path. Hence, the following ants prefer to choose the path with higher pheromone [58]. The ACO algorithm has also been used to find both the optimal assembly and disassembly sequence.

3.1 Tour construction

Since parallel disassembly contains more than one manipulator, ant groups are used to visit the nodes. The size of the ant group equals to the number of the manipulators that involved in parallel disassembly. Let each ant in a group denote a manipulator, and search parallel disassembly sequence cooperatively. Furthermore, to guarantee that the ant colony can cover the solution spaces as much as possible, multiple groups will be sent to search the solutions separately in every round of

iteration. The quantity of groups is determined according to the number of initial detachable parts.

To ensure the geometric feasibility of parallel disassembly sequence, the ants must cooperate with each other. Therefore, all the ants in a group share the same disassembly matrix and disassembly work area collision matrix to guarantee the feasibility of the disassembly sequence. Each ant randomly selects a part from the initial detachable parts and begins to search.

After every ant chooses an initial detachable part, all these parts have been disassembled, and the DM and WACM (the row and column to which belongs to these initial detachable parts chosen by ants will be set to zeros) are updated. Subsequently, based on Formulas (1) and (2) given in Section 2.2, all the feasible parts can be acquired for DM and WACM, which are put into a list $A_k(s)$. Then, all the ants in the group choose the next part that disassembled from $A_k(s)$, as the next node to visit. The transition probability that each part in $A_k(s)$ with all the ants are calculated based on the formula (9). The highest transition probability value of the ant and part combination will be chosen as the disassembled task of the next step.

$$P_{k,ij}(s) = \begin{cases} \frac{[\tau_{ij}(s)]^\alpha [\eta_{k,ij}]^\beta}{\sum_{u \in A_k(s)} [\tau_{iu}(s)]^\alpha [\eta_{k,iu}]^\beta}, & j \in A_k(s) \\ 0, & \text{else} \end{cases} \quad (9)$$

α and β denote the important factors that determine the pheromone and heuristic function, respectively.

Let $\tau_{ij}(s)$ denote pheromone concentration between p_j and p_i . $\eta_{k,ij}$ is defined as the visibility information of ant k as follows:

$$\eta_{k,ij} = \frac{(1-\omega) \cdot \lambda_{k,ij} + \omega \cdot \theta_{k,ij}}{q_\eta} \quad (10)$$

Let ω and q_η represent the weight coefficient and adjustment factor. It should be noted that the execution time of the parts does not change with the order of disassembly, and all

parts need to be disassembled. Thus, the execution time is not taking into account as a variable of visibility information.

In the SDSP problem, after a part chosen by an ant as the next step of disassembly task, the part will be put into a taboo list. Therefore, the other ants cannot visit the part repeatedly. Then, if all the nodes have been visited, a possible solution can be acquired from the taboo list. Unlike the SPDP problem, the index of the chosen part in the taboo list should be cautious when building a feasible solution for parallel disassembly sequence with MMAS. Here, Fig. 4 is given as an example to illustrate the method for building the taboo list for PDSP.

As shown in Fig. 4, let $list(t)$ denote the taboo list at time t , and assume that p_j is chosen by manipulator no.2. Since p_j is chosen from the list $A_k(s)$, therefore, p_j must not have a collision with the part already disassembled by no.2 itself. Thus, the non-collision between p_j and those part already disassembled by other manipulators need to be guaranteed. For the given example which means those parts in the $list(t)$, which belong to the column of manipulator No.1. Based on the disassembly matrix and area collision matrix, by checking the interference relationship one by one between p_j and those parts mentioned above from bottom to top, the index of p_j can be determined when the interference takes place. The generation of the initial disassembly sequence is illustrated by utilizing the flowchart given below:

- Step 0: Get initial detachable part from DM and WACM.
- Step 1: Let every ant chooses an initial detachable part randomly.
- Step 2: Update the local pheromone.
- Step 3: Update the DM and WACM.
- Step 4: Build $A_k(s)$ with the detachable parts that acquired from DM and WACM.
- Step 5: If $A_k(s)$ is empty, go to Step 11, or else continue.
- Step 6: Calculate the transition probability of each part in $A_k(s)$ by Formula (9).
- Step 7: Let every ant choose a part based on the best result calculated by Step 6.
- Step 8: Put the chosen part into the taboo list, and calculate the index of a part in the $list(t)$ using the method shown in Fig. 4.
- Step 9: Calculate the disassembly time using the method given in Section 3.2, and go to Step 3.
- Step 10: Finish; an initial disassembly sequence can be acquired from $list(t)$.

3.2 Waiting strategy based on disassembly time

Note the part that contains in the list $A_k(s)$, which can be disassembled based on the assumption that every manipulator has already finished their task. Therefore, if the parallel disassembly is asynchronous, the selected part may need to wait

until the other manipulators finish the previous tasks. It is noteworthy that without the assumption that every manipulator finishes their tasks, a manipulator can only choose the parts without waiting for the other manipulators to finish their present tasks, and it may lose the opportunity to get the better sequence such as the instance mentioned in Section 2.1. Thus, the waiting strategy needs to be designed to decide whether an ant should wait in each step of the research. The key to waiting strategy is to determine the disassembly priority between the selected parts and the parts that are being performed at the time by the other manipulators.

The incident of waiting takes place, waiting time calculation, and comparison with preparation time are illustrated by utilized the flowchart given below:

- Step 0: Label each part processing at present, and build the current disassembled parts C_k .
- Step 1: Let p_j denote the part that ant k chose from $A_k(s)$.
- Step 2: Check the disassembly and work area interference relationship between p_j and the parts contained in C_k .
- Step 3: If the interference takes place, calculate the current disassembly time that every ant already consumed based on Formula (6) and C_k ; otherwise, go to Step 8.
- Step 4: Compare the current disassembly time, if there is a part finish after ant k , indicating that there is a possibility that before performs the task of disassembly p_j the ant k has to wait for another ant to finish their tasks. Note that only those ants which task interference with ant k participate in the comparison; Otherwise, go to Step 8.
- Step 5: Calculate the difference of current disassembly time between the ant k with other manipulators as the waiting time $\delta_{k, ij}$. If there are multiple interferences take place, the maximum waiting time will be taken as the waiting time of ant k .
- Step 6: Calculate the preparation time that ant k will be spent on p_j by using Formulas (4) and (5) proposed in Section 2.2.
- Step 7: Compare the preparation time $\varepsilon_{k, ij}$ and waiting time $\delta_{k, ij}$. If $\delta_{k, ij} > \varepsilon_{k, ij}$, then ant k will wait for the other manipulators to finish based on Formula (3). The disassembly time $t_{k, j}$ that ant k consumes on p_j is $\sigma_{k, j} + \delta_{k, ij}$; otherwise, go to Step 8.
- Step 8: It is no necessary to wait for other ants the disassembly time that ant k spends on p_j is the sum of execution time and preparation time, therefore $t_{k, j} = \sigma_{k, j} + \varepsilon_{k, ij}$.

3.3 Update of pheromone

To prevent the ant colony to converge to a certain part of the feasible domain during the research, the ants are guided to search toward the optimal direction. After all the ants complete a search round, the pheromone matrix will be updated.

Pheromone enhancement is performed on these positions that contain in the iterative best solution t^{ib} . The pheromone global update calculation formula at all positions is presented as follows:

$$\tau_{i,j}(t+1) = \rho \cdot \tau_{i,j}(t) + \Delta\tau^{best_{ij}} \quad (11)$$

$$\Delta\tau^{best_{ij}} = 1/f(t^{ib}) \quad (12)$$

In the above formula, ρ denotes the pheromone global volatilization rate; τ_0 denotes initial pheromone concentration. Moreover, the min and max value of pheromones is limited: $[\tau_{min}, \tau_{max}]$. If $\tau_{ij} \leq \tau_{min}$, set $\tau_{ij} = \tau_{min}$; if $\tau_{ij} > \tau_{max}$, set $\tau_{ij} = \tau_{max}$. The value of τ_{min} and τ_{max} are in Formula (13) and Formula (14):

$$\tau_{max} = \frac{1}{1-\rho} \cdot \frac{1}{f(t^{ib})} \quad (13)$$

$$\tau_{max} = \frac{\tau_{max}}{2n} \quad (14)$$

3.4 Process of the proposed algorithm

Step 0: Initialize the pheromone and the parameters.

Step 1: Calculate the disassembly position that switches time between parts with Formula (5).

Step 2: Acquire the initial removable part with DM and WACM, as the start nodes of ants. Add the part into the taboo list.

Step 3: Each ant in the ant group will choose an initial removable part, randomly starting the search.

Step 4: Update the DM and WACM, and then acquire the removable part to allow the disassembled list of next step.

Step 5: Each ant calculate the transition probability with the parts in the disassembled list with Eq. (9);

Step 6: Let the ant with the highest transition probability in the ant group execute the disassembly and move a step.

Step 7: Update local pheromone for MMAS with Formula (11).

Step 8: Calculate the chosen part index in the taboo list with the method given in Section 3.1, and update the taboo list.

Step 9: Calculate the disassembly time of the chosen part with the method given in Section 3.2.

Step 10: Update the disassembly list; if the disassembly list is empty, continue, and go to Step 5.

Step 11: If all of the ant group finish a round of search, continue, and go to Step 3.

Step 12: Acquire the best solution in this iteration.

Step 13: Update pheromone for MMAS and ACS with Formula (11).

Step 14: If iteration time reaches to the maximum number of times, stop the algorithm and output the optimal solution.

4 Application and discussion

This section assesses the performance of the proposed parallel disassembly sequence planning algorithm by running experiments. The algorithm is implemented in MATLAB 2014b and executed on an Intel(R) Core(TM) i7 CPU (2.53 GHz/8.00-GB RAM) PC with a Windows 7 operating system.

Its parameter values are as follows:

- 1) By a large number of experiments (shown in Section 4.3) and instruction by the literature [57, 59], the initial parameters of the proposed algorithm are determined as $\alpha = 2$, $\beta = 2$, $\tau_0 = 1$, $\rho = 0.9$, $q_0 = 0.35$.
- 2) The number of the ant group that was sent to the search solutions in iterations is 1.5 times as the number of parts the assembly contains.
- 3) The size of an ant group equals to the number of manipulators that participate in the disassembly operation.

4.1 Application

To verify the parallel disassembly sequence planning algorithm proposed in this paper, a gear reducer with 37 parts is attempted. The product, as shown in Fig. 5, is adopted as a study case. The disassembly matrix P , work area collision matrix W , and contact matrix C are shown in Figs. 9, 10, and 11 of the Appendix, respectively. The disassembly time of 37 parts is given in Table 2. The first column lists the execution time and the second, and the third lists is preparation time which can be divided into tools preparation time and disassembly position switching time. The execution time and tools' preparation time were acquired from the practical operation, similar with the literatures [60, 61]. The disassembly position switching time was acquired from disassembly simulation by Formula (5). For instance, in line 3, according to Fig. 5, part 26 was disassembled by manipulator no.1, the execution time can be acquired from Table 5 of the Appendix. The preparation time is acquired separately; first, based on Formula (4), the tools' preparation time is saved because the disassembly tool is the same as the previous part (part 25); second, the disassembly position switching time is calculated by Formula (5); the bounding box center of part 25 and part 26 is given in Table 6 of the Appendix. (Table 7)

The results from the three repeated tests are given in Table 2. With 200 iterations, the best, worst, and average disassembly times are 258.6640 s, 272.1372 s, and 279.8163

s, respectively; which is much shorter than the 695 s of processing time required by the SDSP method. The Gantt chart of the best solution is shown in Fig. 6. Parts 1, 2, and 3 represent manipulator no.1, no.2, and no.3 in Table 2, correspondingly.

4.2 Discussion

This subsection provides a comparison between the effect of strategy that the preparing time overlaps with the waiting time. After shutting down, the overlapping strategy and the best, worst, and average disassembly time are given in Table 3, when the same parameters are used in Section 4.1. Compared to the result with an overlap strategy mentioned above and given in Table 3, the disassembly time is increasing. Therefore, the subdivided disassembly time can shorten the parallel disassembly time.

To verify the feasibility and superiority of the proposed algorithm, this subsection provides a comparison between the proposed method and the improved GA algorithm given in the literature [12], which design to solve asynchronous PDP problem. An engine block mentioned in literature [31] and a valve taken from [12] are used as test products. The results are given in Table 4. To be consistent between the three approaches, the disassembly time given in the literature [12] is divided into three parts according to the ratio of 30%, 40%, and 30%, which correspond to disassembly execution time, disassembly tool preparation time, and disassembly position switch time, respectively. For the engine block product, the work area collision is not considered, because the work area collision for all the parts is not available in the literature [31]. In addition, the proposed method is implemented 10 times for each instance to have sufficient statistical data for comparison since the algorithm has probabilistic and randomized features. The best disassembly time acquired by each instance is given in Table 4.

From Table 4, the disassembly times of the proposed algorithm are 349.5 s and 1381 s for the valve cover and the engine block, respectively. Both results are better than that of the other approach (i.e., 350.5 s and 1728 s). It should be stressed that the method [12] has not considered the waiting strategy and time overlap in the disassembly processes, suggesting that the two products spent more disassembly time. Therefore, the proposed algorithm can successfully solve disassembly examples, and the quality of the solution results is high. Since large-scale disassembly processes are more common in the industry, the proposed approach is of more practical value.

4.3 Parametric analysis

In this study, the value of key parameters α , β and the number of the ant groups have been tested repeatedly. The parameters tested were examined one by one while others were fixed.

Take the case above as an example: the pheromone evaporation rate $\rho = 0.9$ and the pseudorandom ratio $q_0 = 0.35$. The number of ants group is 56 (1.5 times to the part number of the study case). The number of iterations are 200, as the result shown in Fig. 7; when $\alpha = 2$ and $\beta = 1, 2, 3.5$, the proposed algorithm can acquire a better solution than the other combinations.

Furthermore, the effects of the ant group size on the proposed method are studied, as shown in Figs. 8, which are obtained that the ant group size is 0.5, 1, 1.5, 2 times as the part number of the product, respectively. The maximum number of iterations was 200, the number of manipulators was 3, and the system ran continuously 3 times for every ant group size. The best solution can be acquired by all of them except that the ant group size equals to the part number of product. Considered the time-consuming, the appropriate ant group size and number of iterations should be selected by various factors, such as productivity and quality.

The experiment of the relationship between the number of manipulators and the total disassembly time is also been carried out. When the number of manipulators was larger than 3, i.e., the total disassembly time converged to 248 s, only 24 s has been saved. When the number of manipulators was two, the average of the total disassembly time of 3 tests is 354.4 s, which is much longer than the time spend by three manipulators. Therefore, in actual disassembly work, a greater number of manipulators are not better. The appropriate number of operators should be allocated when the venue and tools allow it.

5 Conclusions

This work proposes a subdivided disassembly time model for dealing with asynchronous parallel disassembly planning problem. Based on MMAS, an efficient heuristic algorithm is adopted to identify the disassembly sequence and manipulators allocation that minimizes the total disassembly time. To deal with the precedence relationship and work area collision, a new disassembly task assign method with waiting strategy that allows a group of ants search feasible sequence solutions cooperatively is developed. The advantage of subdivided disassembly time model is discussed. By using the strategy that makes the preparation time and waiting time overlapped, the waiting time was reduced. The effectiveness of the proposed method is verified by using a gear reduce contains 37 parts. Furthermore, the impact on the improved MMAS algorithm performance by the parameter and the size of ant group is tested and discussed. Results show that the proposed algorithm can find the optimal/near optimal solution in reasonable time.

Appendix

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	
$P=$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	
	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	
	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	
	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	
	1	1	1	1	0	0	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	0	0	0	0	0	0	0	10	
	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	15	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	17	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	19	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	21	
	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	22	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	23	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	1	0	0	0	0	24	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	25	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	27	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	28	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	29	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	30	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	31	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	32	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	33	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	34	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	35	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	36	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	37	

Fig. 9 The disassembly matrix of the transmission in Fig. 4

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37		
C=	0	0	0	0	2	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
	0	0	0	0	2	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
	0	0	0	0	2	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3		
	0	0	0	0	2	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4		
	2	2	2	2	0	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5		
	0	0	0	0	2	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6		
	0	0	0	0	4	0	2	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7		
	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8		
	0	0	0	0	0	2	2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	9		
	2	2	2	2	1	0	0	0	0	0	1	2	2	2	2	0	1	2	2	2	2	0	0	1	2	2	2	2	2	0	0	0	1	2	2	2	2	10	
	0	0	0	0	0	0	0	0	0	1	0	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
	0	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	
	0	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13	
	0	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14	
	0	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	15	
	0	0	0	0	0	0	2	0	1	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	
	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	2	0	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	17	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	19	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	21	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	0	0	0	0	0	0	22	
	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	2	2	0	0	0	0	0	23	
	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	2	2	2	0	0	0	1	2	2	0	1	2	2	24
	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	25	
	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	27	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	28	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	29	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0	0	2	0	0	0	0	0	0	30	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	2	2	0	2	0	2	0	0	0	31	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0	2	0	0	0	32	
	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	2	0	2	2	2	2	2	2	33	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	34	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	35	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	36	
	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	37	

Fig. 10 The contact matrix of the transmission in Fig. 4

[illegible]

Fig. 11 The work area collision matrix of the transmission in Fig. 4

Table 5 The execution time of parts

Part index	No. 1	No. 2	No. 3
1	8.9	8.9	8.9
2	8.9	8.9	8.9
3	8.9	8.9	8.9
4	8.9	8.9	8.9
5	11	11	11
6	40	40	40
7	15	15	15
8	4	4	4
9	13	13	13
10	19	19	19
11	6	6	6
12	5.3	5.3	5.3
13	5.3	5.3	5.3
14	5.3	5.3	5.3
15	5.3	5.3	5.3
16	40	40	40
17	6	6	6
18	8.2	8.2	8.2
19	8.2	8.2	8.2
20	8.2	8.2	8.2
21	8.2	8.2	8.2
22	4	4	4
23	13	13	13
24	10	10	10
25	9.7	9.7	9.7
26	9.7	9.7	9.7
27	9.7	9.7	9.7
28	9.7	9.7	9.7
29	40	40	40
30	8	8	8
31	15	15	15
32	40	40	40
33	9	9	9
34	7.5	7.5	7.5
35	7.5	7.5	7.5
36	7.5	7.5	7.5
37	7.5	7.5	7.5

Table 6 The bounding box center of parts

	X	Y	Z
1	-123.61	196.22	-61.42
2	-123.61	79.55	-61.42
3	-123.61	79.55	55.26
4	-123.61	196.22	55.26
5	-121.61	137.89	-3.08
6	-111.38	138.13	-3.25
7	-54.5	138.13	-3.25
8	68	162.13	-3.25
9	67.75	138.28	-3.21
10	-1.11	125.39	-3.08
11	-1.11	259.89	-3.08
12	-59.45	256.39	-61.42
13	57.23	256.39	-61.42
14	57.23	256.39	55.25
15	-59.45	256.39	55.25
16	108.13	138.13	-3.25
17	119.39	137.89	-3.08
18	121.39	196.22	-61.42
19	121.39	196.22	55.25
20	121.39	79.55	55.25
21	121.39	79.55	-61.42
22	-0.32	138.6	34
23	-0.32	138.42	62.42
24	-0.32	138.6	167.42
25	-71.38	67.53	121.42
26	70.75	67.53	121.42
27	70.75	209.66	121.42
28	-71.38	209.66	121.42
29	-0.32	138.6	100.54
30	-0.32	161.6	62.92
31	-0.32	138.6	191.67
32	-0.32	138.6	215.29
33	-0.32	138.6	249.92
34	-51.58	87.33	248.42
35	50.95	87.33	248.42
36	50.95	189.86	248.42
37	-51.58	189.86	248.42

Table 7 The tool preparation time of parts

Part index	No. 1	No. 2	No. 3
1	5.2	5.2	5.2
2	5.2	5.2	5.2
3	5.2	5.2	5.2
4	5.2	5.2	5.2
5	1.5	1.5	1.5
6	15	15	15
7	2	2	2
8	3	3	3
9	2	2	2
10	10	10	10
11	1.5	1.5	1.5
12	7	7	7
13	7	7	7
14	7	7	7
15	7	7	7
16	15	15	15
17	1.5	1.5	1.5
18	3.8	3.8	3.8
19	3.8	3.8	3.8
20	3.8	3.8	3.8
21	3.8	3.8	3.8
22	8	8	8
23	2	2	2
24	5.5	5.5	5.5
25	6.3	6.3	6.3
26	6.3	6.3	6.3
27	6.3	6.3	6.3
28	6.3	6.3	6.3
29	15	15	15
30	3	3	3
31	2	2	2
32	15	15	15
33	1.5	1.5	1.5
34	4.5	4.5	4.5
35	4.5	4.5	4.5
36	4.5	4.5	4.5
37	4.5	4.5	4.5

Authors' contributions All the authors designed research, performed research, analyzed data, and wrote the paper.

Funding This research is supported by the Key Laboratory of Fundamental Science for National Defence of Aeronautical Digital Manufacturing Process of Shenyang Aerospace University (SHSYS201805).

Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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